ADOMAS VALANTINAS, PHD

Senior Fellow, Leverhulme Centre for Life in the Universe, University of Cambridge, UK
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RESEARCH INTERESTS

I bridge orbital remote sensing and laboratory geochemistry to explore the crustal evolution and habitability of rocky planets.

Topics: Mars, early climate and habitability, iron mineralogy & redox chemistry, planetary materials, crustal & magmatic evolution, aqueous & igneous mineralogy, seasonal processes (frost, dust, mass wasting), impact cratering, reflectance spectroscopy, terrestrial analogue field studies, machine learning, biosignatures, lunar geology, Phobos & Deimos.

RESEARCH APPOINTMENTS

LCLU Senior Fellow Sep 2026 – present
Leverhulme Centre for Life in the Universe (LCLU), University of Cambridge, UK
Host: Prof. Nicholas Tosca
Focus: experimental aqueous geochemistry of iron minerals and potential biosignatures on Mars

SNSF Postdoc Mobility Fellow Sep 2023 – Aug 2026
Brown University, USA (two years) and *ETH Zurich, Switzerland* (one year)
Advisors: Prof. Jack Mustard (Brown) and Prof. Cara Magnabosco (ETH)
Focus: remote sensing and machine learning of aqueous alteration on ancient Mars

Postdoctoral Researcher Dec 2022 – Mar 2023
University of Bern, Switzerland
Advisor: Dr. Antoine Pommerol
Focus: laboratory reflectance spectroscopy of iron oxides

Graduate Research Assistant (Ph.D.) Oct 2018 – Nov 2022
University of Bern, Switzerland
Advisor: Prof. Nicolas Thomas
Focus: present-day climate of Mars and related surface processes: dust, frost and mass wasting

Master's Degree Researcher Oct 2015 – Oct 2017
University of Copenhagen, Denmark
Advisor: Prof. Kjartan Kinch
Focus: development and characterization of the Mastcam-Z flight calibration targets now operating on NASA's Perseverance rover

Student Research Assistant (software) Oct 2015 – Oct 2017
Microsoft Station Q and Center for Quantum Devices, Niels Bohr Institute, University of Copenhagen, Denmark
Focus: software development for laboratory instrument control and computer hardware maintenance

EDUCATION

Ph.D. Astronomy and Physics Oct 2018 – Nov 2022
University of Bern, Switzerland
Thesis: *Bright and Dusty Regions of Mars: New Insights from Experiments and Orbital Color Imaging*
Advisor: Prof. Nicolas Thomas

M.Sc. Geophysics Oct 2015 – Oct 2017
University of Copenhagen, Denmark
Advisor: Prof. Kjartan Kinch

B.Sc. Astronomy and Physics Sep 2011 – Jul 2015
Vilnius University, Lithuania
Advisor: Dr. Audrius Bridzius

Astronomy and Physics (year abroad) Sep 2011 – Jun 2012
University of Glasgow, UK

GRANTS AND AWARDS

AI for Science Program award, Anthropic (PI; $20,000 in Claude API credits) October 2026
LCLU Senior Fellowship, Leverhulme Centre for Life in the Universe, University of Cambridge (PI; $374,000) Sep 2026 – Aug 2030
NASA Solar System Science program, "Simulating cloud formation on Mars in the laboratory: probing the onset of nucleation in a realistic Martian atmosphere environment", (Collaborator to PI Dr. Marek Slipski, NASA Jet Propulsion Laboratory; ~$150,000) Apr 2026 – Mar 2027
Swiss National Science Foundation (SNSF) Postdoc Mobility return grant, "Potential for ancient habitability and deep crustal materials preserved in the rim of Jezero crater, Mars", ETH Zurich (PI; $155,000) Sep 2025 – Aug 2026
Swiss National Science Foundation (SNSF) Postdoc Mobility Fellowship, "Windows into Mars' potentially habitable subsurface in the context of NASA's Perseverance extended mission", Brown University (PI; $146,000) Sep 2023 – Aug 2025
American Philosophical Society Baruch Blumberg Grant for Astrobiology, "Decoding Mars Through Iceland: Ferrihydrite as a Key to Past Habitability and Potential Biosignatures", for field work and conference attendance in Iceland (PI; $8,500) July 2025
CIFAR seed funding, "Interfaces of Possibility: Modeling Boundary Ecosystems Across Planetary Systems" (Co-PI; $7,200) May 2025
Lunar and Planetary Institute (LPI) Career Development Award (meeting canceled due to COVID) March 2020
American Geophysical Union (AGU) student travel award ($2,000) December 2017
Oticon Foundation (Oticon Fonden) grant, Denmark (~$3,300) June 2016

MISSION EXPERIENCE

NASA Perseverance Rover, PIXL Science Team collaborator (PI: A. Allwood). Contributing to science discussions and operations planning. Oct 2026 – present

ESA Trace Gas Orbiter (TGO) / CaSSIS, Science Team collaborator and operations planner (PI: A. Pommerol). Planned 10 months of CaSSIS observations (~10% of total mission time). Science Theme lead for Hydrothermal Processes and for Geologic Contacts & Stratigraphy. Contributed 31,000 targets to the CaSSIS targeting database (~60% of all team targets). Oct 2018 – present

NASA Mars Reconnaissance Orbiter (MRO) / HiRISE, Science Team collaborator (PI: S. Byrne). Regular contributor to the HiRISE targeting database. Jun 2020 – present

NASA Double Asteroid Redirection Test (DART), guest participant in the DART Boarders program (PI: A. Cheng). Observed team meetings and took part in discussions. Sep 2022 – Jan 2023

NASA Perseverance Rover, Mastcam-Z Science Team collaborator (PI: J. Bell). Developed and characterized the flight calibration targets now operating on Mars. Oct 2015 – Oct 2017

RESEARCH METHODS AND TECHNICAL EXPERTISE

Laboratory spectroscopy: VNIR–SWIR reflectance spectroscopy and spectro-goniometry (analogue soils, dust, frosts); FTIR; Hapke radiative-transfer and intimate-mixture modelling.

Wet-lab: Fe-(oxyhydr)oxide synthesis; planetary regolith analogue synthesis; XRD; experimental aqueous geochemistry of Fe minerals.

Orbital remote sensing: Visible imaging (0.4–1.1 µm) — HiRISE/MRO, CTX/MRO, CaSSIS/TGO, HRSC/MEx, MOC/MGS, LROC/LRO. Near-infrared spectroscopy (0.4–5 µm) — CRISM/MRO, OMEGA/MEx, M³/Chandrayaan-1, IIRS/Chandrayaan-2, SP/Kaguya. Thermal infrared (6–50 µm) — THEMIS/Odyssey, TES/MGS, Diviner/LRO. Topography and mapping — MOLA/MGS, HiRISE/CTX stereo DEMs (Ames Stereo Pipeline), crater size–frequency dating, geologic mapping.

Machine learning and data science: Unsupervised clustering and dimensionality reduction (k-means, HDBSCAN, UMAP) and supervised classification of CaSSIS and M³ spectra; deep-learning (CNN) object detection in global orbital image catalogues; zero-/few-shot image classification and feature mapping with multimodal LLMs (Claude API).

Flight hardware and operations: Mastcam-Z/Perseverance calibration targets (development, characterization); CaSSIS/TGO and HiRISE/MRO targeting and observation planning; PIXL and SuperCam/Perseverance operations (incoming).

Field work: Volcanic, hydrothermal and Fe-rich spring environments sampled for spectral and mineralogical analysis — Iceland, Azores (PT), Canary Islands (ES), Block Island (RI).

FIRST-AUTHOR PUBLICATIONS

1. **Valantinas, A.**, Rajsic, A., Mustard, J., Tullo, A. et al. Partially altered olivine in the western rim of Jezero crater, Mars: preserved pre-impact Noachian basement. In review at *GRL*; preprint on *ESS Open Archive*. doi:10.22541/essoar.175579625.58968122/v3
2. **Valantinas, A.**, Mustard, J., Tullo, A., Bramble, M.S. and Head, J.W. An exposed magmatic feeder system for the regional olivine unit in Nili Fossae, Mars. *Earth and Planetary Science Letters*, 696, 120390 (2026). doi:10.1016/j.epsl.2026.120390
3. **Valantinas, A.**, Mustard, J., Chevrier, V. et al. Detection of ferrihydrite in Martian red dust records ancient cold and wet conditions on Mars. *Nature Communications*, 16, 1712 (2025). doi:10.1038/s41467-025-56970-z
4. **Valantinas, A.**, Thomas, N., Pommerol, A. et al. Evidence for transient morning water frost deposits on the Tharsis volcanoes of Mars. *Nature Geoscience*, 17, 608–616 (2024), cover article. doi:10.1038/s41561-024-01457-7
5. **Valantinas, A.**, Becerra, P., Pommerol, A. et al. CaSSIS color and multi-angular observations of Martian slope streaks. *Planetary and Space Science*, 209, 105373 (2021). doi:10.1016/j.pss.2021.105373
6. **Valantinas, A.** and Schultz, P.H. The origin of neotectonics on the lunar nearside. *Geology*, 48, 649–653 (2020). doi:10.1130/G47202.1
7. **Valantinas, A.**, Kinch, K.M. and Bridzius, A. Low crater frequencies and low model ages in lunar maria: Recent endogenic activity or degradation effects? *Meteoritics & Planetary Science*, 53, 826–838 (2018). doi:10.1111/maps.13033

CO-AUTHOR PUBLICATIONS

8. Bramble, M., (…) **Valantinas, A**. et al. An intrusive origin for the Nili fossae regional olivine-bearing unit. Accepted at *Communications Earth & Environment*.
9. Munaretto, G., (...) **Valantinas, A**. et al. Evidence of mafic silicates in the blue unit of Phobos: Vis-NIR analogs within the C and Xc-type asteroid taxonomy. *Astronomy & Astrophysics*, in press (2026). doi:10.1051/0004-6361/202661396

10. Beccarelli, J., (...) **Valantinas, A.** et al. Spectral variability of Phobos and Deimos from TGO/CaSSIS multiband observations. *Astronomy & Astrophysics*, 707, L6 (2026). doi:10.1051/0004-6361/202558051

1. Munaretto, G., (...) **Valantinas, A.** Phase reddening of Phobos and Deimos from TGO/CaSSIS observations. *Astronomy & Astrophysics*, 700, L1 (2025). doi:10.1051/0004-6361/202555720

2. Bickel, V. and **Valantinas, A.** Streaks on Martian slopes are dry. *Nature Communications*, 16, 4315 (2025). doi:10.1038/s41467-025-59395-w

3. Thomas, N., (...) **Valantinas, A.** et al. Seasonal and short timescale changes on the Martian surface: Multi-spacecraft perspectives. *Space Science Reviews*, 221, 3 (2025). doi:10.1007/s11214-024-01128-4

4. Mills, M.M., (...) **Valantinas, A.** A global dataset of pitted cones on Mars. *Icarus*, 418, 116145 (2024). doi:10.1016/j.icarus.2024.116145

5. Pommerol, A., (...) **Valantinas, A.** et al. In-flight radiometric calibration of the ExoMars TGO Colour and Stereo Surface Imaging System. *Planetary and Space Science*, 223, 105580 (2022). doi:10.1016/j.pss.2022.105580

6. Munaretto, G., (...) **Valantinas, A.** et al. Multiband photometry of Martian Recurring Slope Lineae (RSL) and dust-removed features at Horowitz crater, Mars from TGO/CaSSIS color observations. *Planetary and Space Science*, 214, 105443 (2022). doi:10.1016/j.pss.2022.105443

7. Dundas, C.M., (...) **Valantinas, A.** et al. Active Mars: A dynamic world. *JGR: Planets*, 126, e2021JE006876 (2021). doi:10.1029/2021JE006876

MANUSCRIPTS IN PREPARATION

8. **Valantinas, A.**, Girjoaba, A. and Magnabosco, C. Acid Mars, not so acidic? Widespread ferrihydrite–sulfate deposits in CRISM VNIR data. Oral presentation at AGU Fall Meeting 2026. Supported by the **Anthropic AI for Science Program**.

9. **Valantinas, A.** and Tosca, N. Abiotic formation of leopard-spot redox features: implications for potential biosignatures on Mars.

10. Baisch, E.J., D'Andria, M., **Valantinas, A.**, Cooper, T.A. and Warren, K.J. Low-temperature water and preparation-dependent oxygen from heating a ferrihydrite analog of Martian dust. Student-led manuscript; co-supervised.

11. Miller, S.G., (...) **Valantinas, A.** et al. TIDES: A Triton Multiple-Flyby Mission Concept for the New Frontiers Program.

STUDENT SUPERVISION

Elia J. Baisch, **M.Sc. Engineering student** (ETH Zurich, Solar Energy Engineering Laboratory). M.Sc. thesis project on water and oxygen release from heating ferrihydrite-bearing Martian dust analogues; first-author manuscript in preparation. Co-supervisor (main supervisor: Prof. Kent J. Warren). — Feb 2026 – present

Luis Grimaldo Estrada, **M.Sc. Data Science student** (ETH Zurich). Machine Learning semester project and M.Sc. thesis project (incoming). Main supervisor. — Fall 2026

Ari Antje Hölscher, **M.Sc. Data Science student** (ETH Zurich). Machine Learning for Earth and Planetary Sciences III summer project. Main supervisor; co-supervised with Prof. Cara Magnabosco. — Summer 2026

Chen Orenshtein, **M.Sc. Data Science student** (ETH Zurich). Machine Learning for Earth and Planetary Sciences III summer project. Main supervisor; co-supervised with Prof. Cara Magnabosco. — Summer 2026

Andrei Girjoaba (ETH Zurich, Computing Platforms - Systems Group), M.Sc.-level **paid research assistant**. Machine learning for Mars remote sensing data. Main supervisor; co-supervised with Prof. Cara Magnabosco.	Mar 2026 – Sep 2026
Guilliat Corboz, **B.Sc. thesis student** (ETH Zurich, D-EAPS). "Searching for Hydrothermal Systems on Mars: Using Polygonal Ridges to Map Subsurface Habitability". Main supervisor; co-supervised with Prof. Cara Magnabosco.	Spring 2026
Rushil Kukreja, high school **research mentee** (Brown). Spectral analysis of Martian orbital data.	Fall 2024

TEACHING

Research mentor, MPhil in Planetary Science and Life in the Universe, University of Cambridge.	Fall 2026
Teaching assistant, *Machine Learning for Earth and Planetary Sciences II & III*, ETH Zurich.	Spring & Summer 2026
Teaching assistant, lab course in *thermodynamics*, University of Bern.	Spring 2022
Teaching assistant, lab course in *optics*, University of Bern.	Fall 2021
Teaching assistant, lab course in *thermodynamics*, University of Bern.	Spring 2021
Teaching assistant, lab course in *optics*, University of Bern.	Fall 2020
Teaching assistant, lab course in *thermodynamics*, University of Bern.	Spring 2020
Private tutor in physics and mathematics for high school students, Lithuania.	2017 – 2018

PROFESSIONAL SERVICE AND LEADERSHIP

Co-convener and session chair, "The Emergence of Life as a Planetary Process: Prebiotic Chemistry and Habitability on the Early Earth and Beyond", AGU Fall Meeting 2026.	December 2026
Organizer, LCLU reading group, University of Cambridge.	Fall 2026
Primary convener and session chair, "The Emergence of Life as a Planetary Process", AGU Fall Meeting 2025.	December 2025
Organizer, Astrobiology and Origin of Life (OLA) reading group, Brown University.	Jan 2024 – May 2025
Organizer, Planetary Lunch Bunch seminar series, Brown University.	Spring 2024
Panel member, NASA Mars Data Analysis Program (MDAP) review panel.	January 2024
Dwornik Award judge for student presentations, Lunar and Planetary Science Conference.	Mar 2023 & Mar 2025
Peer reviewer for *Earth-Science Reviews*, *Geology*, *Geophysical Research Letters*, *Journal of Geophysical Research*, *Planetary Science Journal* and *Icarus*.	Sep 2021 – present
Member, Geological Society of America.	Sep 2021 – present
Member, American Geophysical Union.	Dec 2017 – present

OUTREACH

Public talk, *Astronomy on Tap Vilnius*, Lithuania (~30 attendees).	June 2026
Public talk, *Astronomy on Tap Providence*, Rhode Island, USA (~100 attendees).	April 2025
Invited guest on the Planetary Society's *Planetary Radio* podcast: "Why is Mars red? A new clue to the history of habitability in Martian dust".	April 2025

Radio DJ and science communicator at NTS.live and RadioVilnius.live, curating shows that combine experimental electronic music with planetary science for non-academic audiences. — Jun 2023 – present

Artistic collaboration with Matteo Taramelli for the exhibition "Potentially Uninhabitable Worlds", Dreiviertel Gallery, Bern, Switzerland. — June 2021

SELECTED PRESS COVERAGE

Martian slope streaks are dry (Bickel & Valantinas, *Nature Communications*, 2025). 130 news outlets; Altmetric 1,031 (top 1%). — May 2025
Space.com · BBC · Universe Today · EarthSky · ESA

Why Mars is red: water-rich ferrihydrite in Martian dust (Valantinas et al., *Nature Communications*, 2025). 187 news outlets; Altmetric 1,497 (top 1%; 5th of 2,489 Nat. Commun. articles of similar age). — February 2025
CNN · BBC · NASA · ESA · SETI Institute · Smithsonian · Boston Globe · Royal Society of Chemistry

First detection of water frost on Mars's tallest volcanoes (Valantinas et al., *Nature Geoscience*, 2024; cover article). 311 news outlets; Altmetric 2,403 (10th of 4,065 Nat. Geosci. articles ever tracked). — June 2024
BBC World News (**live TV**) · CNN · The Guardian · New Scientist · ABC News · USA Today · Smithsonian · ABC Australia (**live radio**) · ESA

Recent tectonic activity on the Moon (Valantinas & Schultz, *Geology*, 2020). — May 2020
Space.com · Ciel et Espace · RTS (Swiss radio)

INVITED TALKS AND SEMINARS

Centre for Origin and Prevalence of Life (COPL) seminar, ETH Zurich, Switzerland — March 2026

Harvard–MIT Planetary Science Meeting, Harvard, USA — January 2025

Harvard–MIT Planetary Science Meeting, MIT, USA — September 2024

Planetary Lunch Bunch, Brown University, USA — September 2024

Planetary Lunch Bunch, Brown University, USA — January 2024

Observatoire de la Côte d'Azur, Nice, France — October 2017

SELECTED CONFERENCE ABSTRACTS (FIRST AUTHOR)

Selected from 40+ conference abstracts

Valantinas, A., Girjoaba, A. and Magnabosco, C. Acid Mars, not so acidic? Widespread ferrihydrite–sulfate deposits in CRISM VNIR data. *AGU Fall Meeting 2026,* San Francisco (oral). — December 2026

Valantinas, A. Unraveling the redox history of Mars. *Origins Federation Annual Meeting* 2026 in Switzerland (poster). — September 2026

Valantinas, A., Mustard, J.F., Tullo, A., Bramble, M.S. and Head, J.W. Evidence for intrusive magmatism as the source of the olivine deposit in the Nili Fossae region, Mars. *57th Lunar and Planetary Science Conference*, abstract 1470 (oral). — March 2026

Valantinas, A., Mustard, J.F., Chevrier, V. et al. Unlocking early Mars redox conditions: evidence from planetary-scale iron oxide deposits. *Mars Through Time International Conference*, Paris, France (oral). — October 2025

Valantinas, A., Rajsic, A. and Mustard, J.F. Potential for ancient habitability and deep crustal materials preserved in the rim of Jezero crater. *Biennial European Astrobiology Conference (BEACON)*, Reykjavík, Iceland (oral).	July 2025
Valantinas, A., Rajsic, A. and Mustard, J.F. Potential for ancient habitability and deep crustal materials preserved in the rim of Jezero crater. *56th Lunar and Planetary Science Conference*, abstract 2068 (oral).	March 2025
Valantinas, A. and Mustard, J.F. Ferrihydrite in dust and bedrock on Mars: astrobiological significance. *Astrobiology Science Conference (AbSciCon)*, Providence, Rhode Island (oral).	May 2024
Valantinas, A., Mustard, J.F. and Bickel, V.T. 'Windows' into Mars' potentially habitable subsurface. *AGU Fall Meeting 2023*, San Francisco (oral).	December 2023
Valantinas, A., Mustard, J.F., Chevrier, V. et al. Orbital evidence of ferrihydrite in the Martian dust: implications for the ancient climate on Mars. *Biennial European Astrobiology Conference (BEACON)*, La Palma, Spain (oral).	May 2023

WORKSHOPS AND SUMMER SCHOOLS

NASA Planetary Science Summer School (PSSS) for mission design, JPL, Pasadena, USA. Helped design a concept mission to Neptune's moon Triton.	August 2025
CIFAR and SNSF "Forum on Radical Interdisciplinarity", Geneva, Switzerland.	March 2025
CIFAR "Earth 4D: Subsurface Science & Exploration" program meeting, Quebec City, Canada.	June 2024
European Astrobiology Institute, "Life in the Sub-Surface: Habitats, Species, Metabolism and Survival Strategies", Azores, Portugal.	October 2023
European Space Agency workshop "Mars V", Les Houches, France.	April 2022
GMAP (Geologic Mapping of Planetary Bodies) Planetary Geologic Mapping Winter School (virtual).	January 2022
GeoPlaNet analogue field winter school, "Fluid-Rock Interactions in the Solar System", Nantes, France.	November 2018
Nordic Astrobiology Network field summer school, "Impacts and their Role in the Evolution of Life", Saaremaa, Estonia.	July 2017
Nordic Astrobiology Network field summer school, "Biosignatures and the Search for Life on Mars", Iceland.	June 2016

FIELD EXPERIENCE

Iceland: field project funded by the Baruch Blumberg Grant for Astrobiology.	July 2025
Block Island, USA: characterization of iron-rich springs (Brown University field trip).	March 2024
Azores, Portugal (Pico, Faial, São Miguel and Terceira): sampling of volcanic and hydrothermal environments, funded by SNSF. Ferrihydrite samples from Terceira were published in *Nature Communications* in 2025.	November 2023
Iceland: adenosine triphosphate (ATP) analysis and collection of volcanic tephra, Nordic Astrobiology Network summer school.	June 2016